1.INTRODUCTION

A)Project Overview

The Strategic Product Placement Analysis is an advanced analytical framework designed to quantify how the physical location of a product within a retail environment directly influences consumer purchasing behavior. By leveraging Tableau Visualization, this project transforms raw transactional data into a strategic tool that "unveils" hidden sales patterns and provides actionable insights for shelf-space optimization**.**

B) purpose

Quantify Revenue Lift: Measure the exact sales increase when moving products to "Prime" zones like eye-level or end-caps.

Map Customer Behavior: Align the customer's in-store journey with transaction data to see how placement drives choice.

Integrate Data Silos: Merge planogram files from cloud storage with live POS database transactions for a unified view.

Enable Field Action: Deliver mobile-responsive Tableau dashboards so managers can make "Keep or Move" decisions on the spot.

2.Ideation Phase Define the Problem Statements

Customer Problem Statement Template:

1. The Customer

Who is the customer?

The primary customers are Retail Store Managers, Category Analysts, and Merchandising Directors who manage physical store layouts.
What is the customer's goal?
To maximize revenue per square foot by ensuring high-margin products are placed in the most visible and accessible locations for shoppers.

2. The Problem

What is the problem?

There is a significant "Visibility Gap." Stakeholders know what is selling, but they don't have a visual way to see where it is selling best within the store. When does the problem occur?

This occurs during Shelf Planning (Plannogramming) and Monthly Sales Reviews when managers try to figure out why certain high-quality products are underperforming.

Where does the problem occur?

On the retail floor—specifically in "Dead Zones" (corners or bottom shelves) that receive high foot traffic but low sales conversion.

3. The Impact

How does it affect the customer?

Without spatial data, managers make "gut-feeling" decisions. This leads to Revenue Leakage, where expensive floor space is wasted on low-margin items while premium products remain hidden.

What is the negative outcome?

Lower "Average Basket Value" because customers aren't being guided toward complementary or impulse-buy products through strategic placement.

4. The Ideal State

What is the desired outcome?

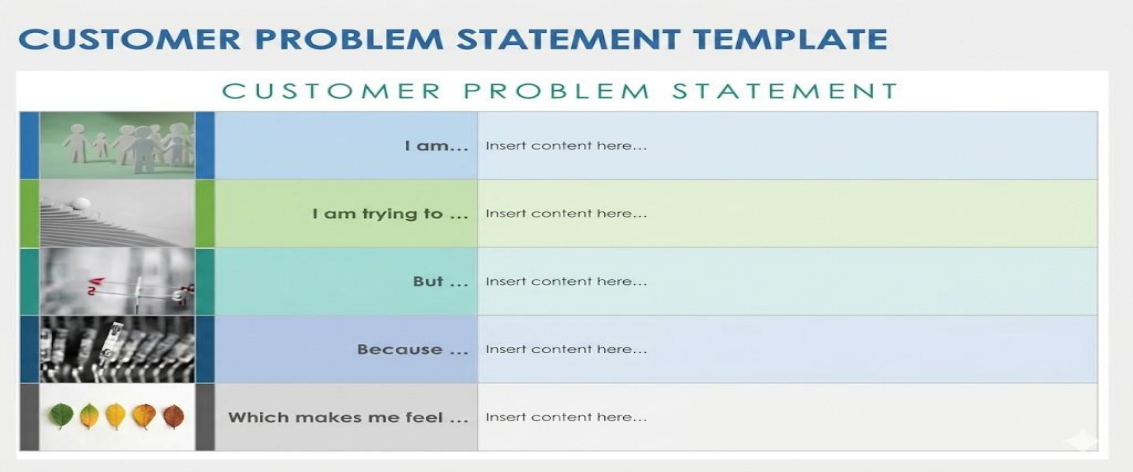
An Interactive Tableau Dashboard that overlays transaction data onto a 2D store map.

How will this help?

Managers will be able to "see" the sales impact of moving a product from a bottom shelf to eye level, allowing them to transform low-performing aisles into high-profit zones.

5. Problem Statement Summary

"Retail Managers are currently struggling to quantify the financial impact of product positioning. Because they lack visual analytics, they are unable to identify 'Dead Zones' on the floor. Our project solves this by using Tableau Visualization to map sales data to physical locations, enabling data-driven placement that increases overall store profitability."



Component Project-Specific Matter

I **am a**: Retail Store Manager or Category Analyst responsible for maximizing store revenue and managing shelf inventory.

I'm trying to optimise product placement on shelves and within store zones to increase sales of high-margin items.

But I lack visual data to see which specific locations (eye-level vs. bottom shelf) or store zones are "dead zones" vs. "hot spots".

Because current sales reports only show what sold, not where it sold, making it difficult to quantify the spatial impact on consumer behavior.

Which makes me feel Frustrated and uncertain about whether the current store layout is actually maximizing potential profit.

2.Ideation Phase Empathize & Discover

Empathy Map Canvas:

Empathy Map: Retail Manager

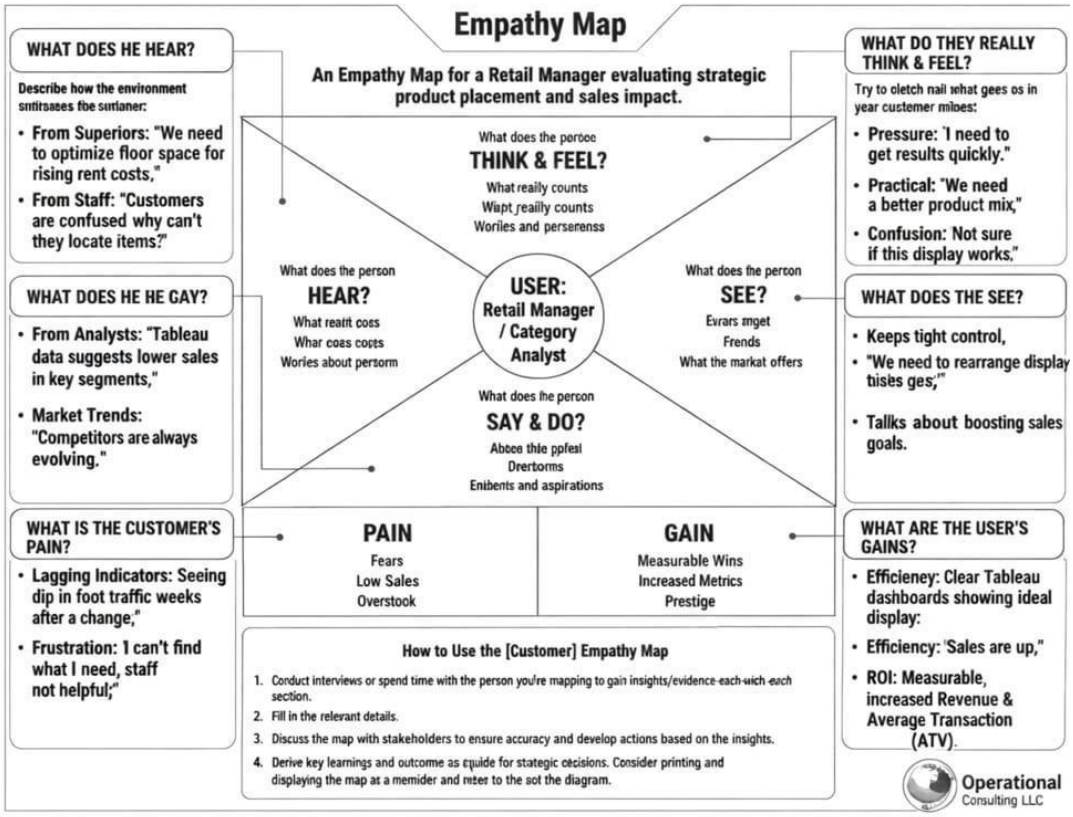
1. WHO are we empathizing with?
   - User: Retail Store Manager / Category Analyst.
   - Goal: To maximize sales by placing the right products in the right spots.
2. What do they SEE?

   - Complicated spreadsheets with rows of numbers.

- "Dead zones" in the store where customers don't stop.
- Competitors using advanced heat-mapping technology.

3. What do they SAY?
   - "I think the bottom shelf is killing our sales for this brand."
   - "We need a visual way to see our floor performance."
   - "Why is this high-margin item not moving?"
4. What do they DO?
   - Manually move products based on "gut feeling."
   - Spend hours trying to cross-reference sales with floor maps. ● Ask staff which aisles seem the busiest.
5. What do they HEAR?
   - From Boss: "Revenue per square foot must increase by 10%."
   - From Staff: "Customers are complaining they can’t find the promotions."
   - From Tech: "Tableau can solve this if we have the right data."
6. What do they FEEL? (Pains & Gains)
   - PAIN: Frustrated by "invisible" sales losses they can't prove.
   - GAIN: Excited to finally see a Heat Map that shows exactly where the money is.

Example:



3. Ideation Phase Brainstorm & Idea Prioritization Template

PROJECT COMPONENT: **BRAINSTORM & IDEA PRIORITIZATION**

Project Name: Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

Objective: To identify and prioritize data-driven strategies that optimize product positioning to maximize retail revenue.

1. Conceptual Brainstorming

Before building the dashboard, we categorized potential factors that influence consumer behavior into four key areas:

- Spatial Mapping: How the physical layout (Aisle vs. Entrance) dictates foot traffic.

- Vertical Hierarchy: The impact of shelf height (Eye-level vs. Reach-level) on brand conversion.
- Product Affinity: The correlation between neighboring products (Cross-merchandising).
- Temporal Dynamics: How placement effectiveness changes based on peak shopping hours.

2. Idea Prioritization Framework

This framework serves as the roadmap for our analysis, categorizing ideas by their strategic value and implementation priority.

Category: Zone Profitability

- **Description:** Mapping revenue to a physical 2D store layout.
- **Strategic Value:** High. It identifies "Hot" spots (high sales) and "Cold" spots (dead zones).
- **Priority:** Critical (P1).

Category: Shelf-Level Lift

- **Description:** Calculating the percentage growth in sales when a product is moved to eye-level.
- **Strategic Value:** High. It optimizes the shelf stocking strategy for high-margin goods.
- **Priority:** Critical (P1).

Category: Market Basket Analysis

- **Description:** Analyzing if placement near complementary items increases the total order size.
- **Strategic Value:** Medium. It helps increase the Average Order Value (AOV).
- **Priority:** High (P2).

Category: Time-of-Day Impact

- **Description:** Determining if placement matters more during morning rushes versus evening rushes.
- **Strategic Value:** Low. It is harder to execute frequent physical changes to a store layout.
- **Priority:** Low (P3).

3. Implementation Logic

To execute these prioritized ideas, the project will follow this logical flow:

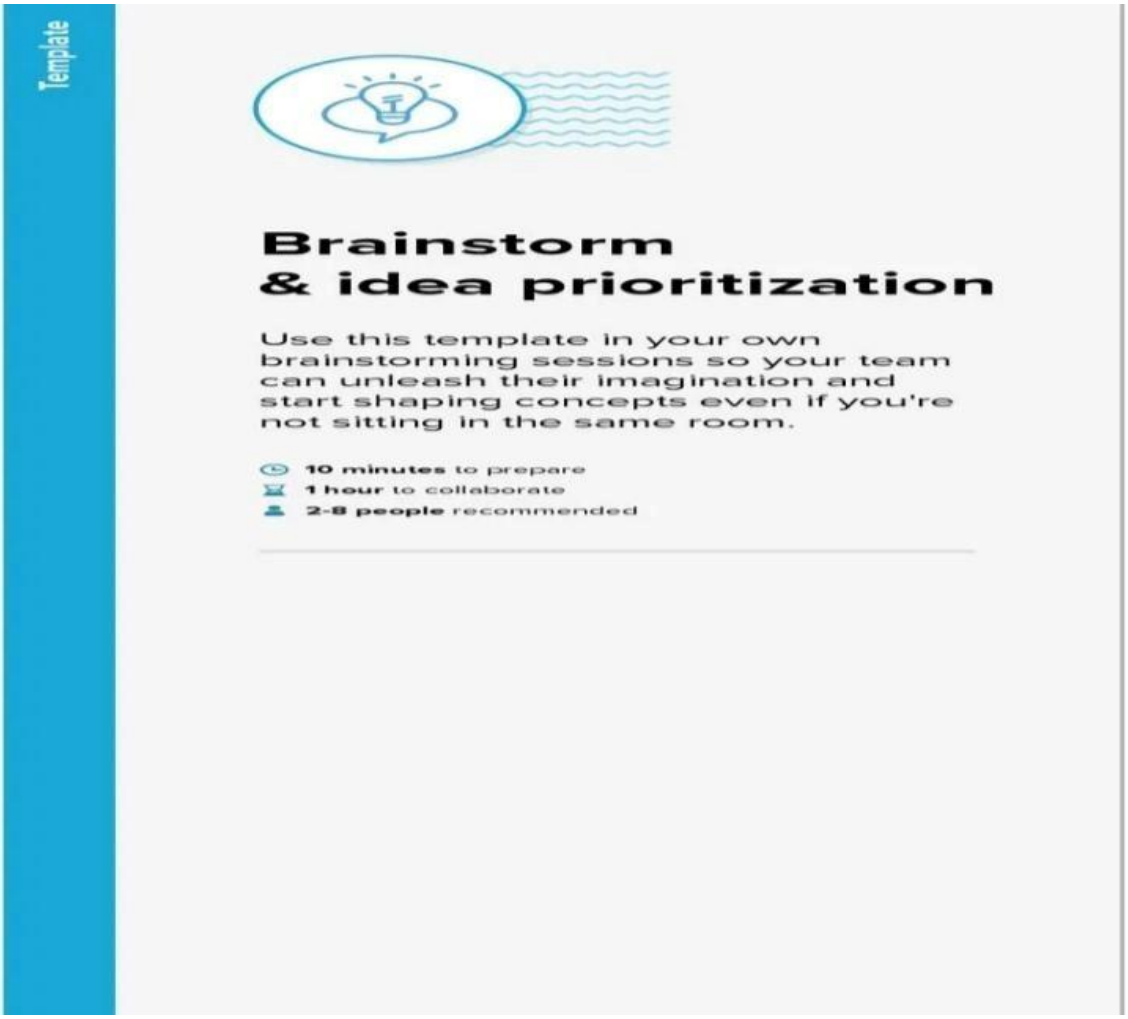
- **For Zone Profitability:** We will use Tableau Spatial Files or custom background images of the store floor plan to create a Heat Map. This allows stakeholders to visualize the "Value of Space" instantly.
- **For Shelf-Level Lift:** We will use Grouped Bar Charts to compare sales across four specific levels: Top Shelf, Eye-Level, Touch-Level, and Bottom Shelf.
- **For Market Basket Analysis:** We will utilize Highlight Tables to show the correlation between the placement of "Product A" and the sales lift of "Product B."

4. Strategic Justification

We have labeled **Zone Profitability** and **Shelf-Level Lift** as **Critical**. Because retail floor space is expensive, understanding the "Revenue per Square Foot" based on exact location provides the fastest Return on Investment (ROI) for store managers and

Step-1:

Team Gathering, Collaboration and Select the Problem Statement



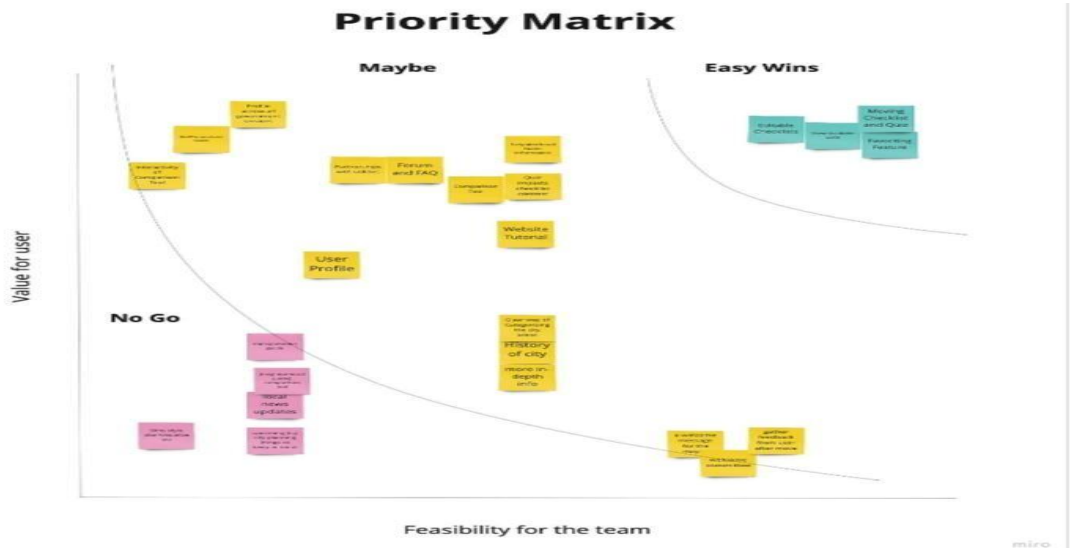
Step-2:

Brainstorm, Idea Listing and Grouping



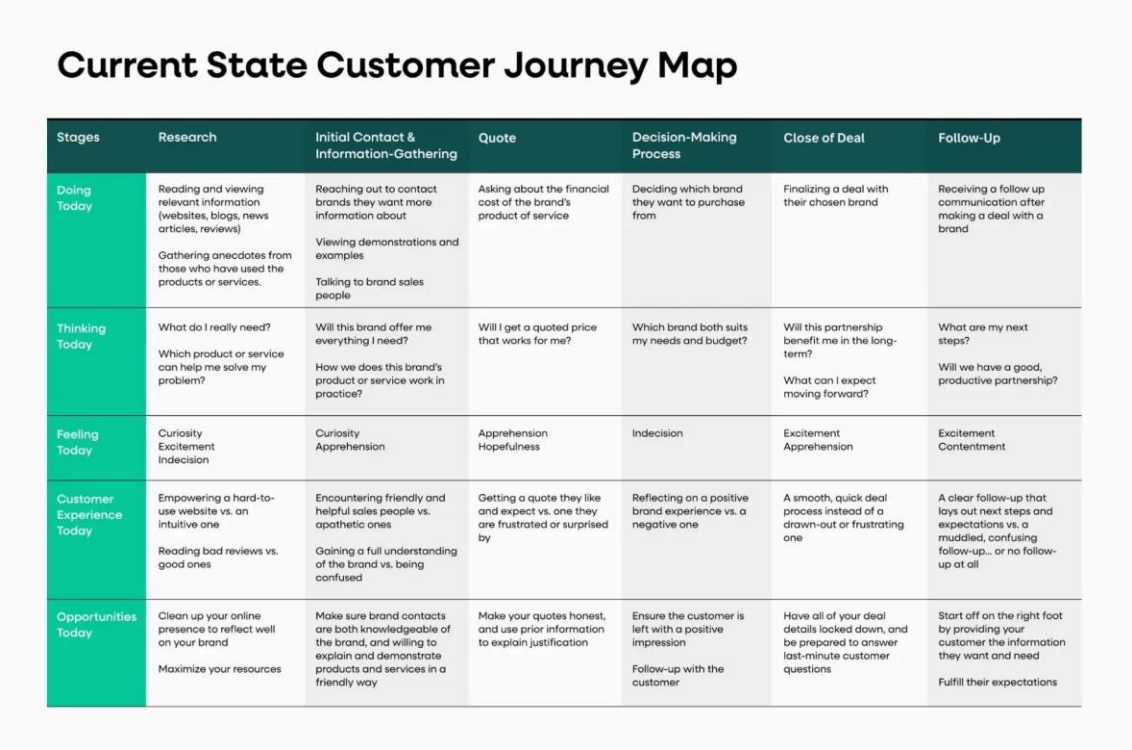
Step-3:

Idea Prioritization



Customer Journey Map

Strategic product placement analysis: unveiling sales impact with tableau visualization



Current State Customer Journey Map

Stages	Research	Initial Contact & Information-Gathering	Quote	Decision-Making Process	Close of Deal	Follow-Up
Doing Today	Reading and viewing relevant information (websites, blogs, news articles, reviews) Gathering anecdotes from those who have used the products or services.	Reaching out to contact brands they want more information about Viewing demonstrations and examples Talking to brand sales people	Asking about the financial cost of the brand's product of service	Deciding which brand they want to purchase from	Finalizing a deal with their chosen brand	Receiving a follow up communication after making a deal with a brand
Thinking Today	What do I really need? Which product or service can help me solve my problem?	Will this brand offer me everything I need? How we does this brand's product or service work in practice?	Will I get a quoted price that works for me?	Which brand both suits my needs and budget?	Will this partnership benefit me in the long-term? What can I expect moving forward?	What are my next steps? Will we have a good, productive partnership?
Feeling Today	Curiosity Excitement Indecision	Curiosity Apprehension	Apprehension Hopefulness	Indecision	Excitement Apprehension	Excitement Contentment
Customer Experience Today	Empowering a hard-to-use website vs. an intuitive one Reading bad reviews vs. good ones	Encountering friendly and helpful sales people vs. apathetic ones Gaining a full understanding of the brand vs. being confused	Getting a quote they like and expect vs. one they are frustrated or surprised by	Reflecting on a positive brand experience vs. a negative one	A smooth, quick deal process instead of a drawn-out or frustrating one	A clear follow-up that lays out next steps and expectations vs. a muddled, confusing follow-up... or no follow-up at all
Opportunities Today	Clean up your online presence to reflect well on your brand Maximize your resources	Make sure brand contacts are both knowledgeable of the brand, and willing to explain and demonstrate products and services in a friendly way	Make your quotes honest, and use prior information to explain justification	Ensure the customer is left with a positive impression Follow-up with the customer	Have all of your deal details locked down, and be prepared to answer last-minute customer questions	Start off on the right foot by providing your customer the information they want and need Fulfill their expectations

4.Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection and Extraction of Data	□ Collect the dataset □ Understanding the Data □ Connecting the dataset with Tableau
FR-2	Data Preparation	□ Prepare the Data for Visualization
FR-3	Data Visualization	□ No of Unique Visualizations
FR-4	Dashboard	□ Responsive and Design of Dashboard

FR-5	Story	No of Scenes of Story
FR-6	Performance Testing	Utilization of Filters No of Calculation Fields
		No of Visualizations/ Graphs
FR-7	Web integration	Dashboard and Story embed with UI With Flask
FR-8	Project Demonstration & Documentation	Record an explanation Video for the project end to end solution Project Documentation-Step by step project development procedure

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No	**Non-Functional Requirement**	**Description**
NFR-1	Usability	The dashboard must feature an intuitive UX, allowing category managers to filter by region, SKU, and shelf position with zero training. It should follow visual best practices (e.g., color-blind friendly palettes).

NFR-2	Security	Data access must be governed by Row-Level Security (RLS), ensuring regional managers only see data for their specific stores. Connection to the data source must be encrypted via SSL/TLS.
NFR-3	Reliability	The system must maintain 99.9% data accuracy between the source ERP/POS system and the Tableau visualization. Automated data quality checks should flag anomalies in sales impact metrics.
NFR-4	Performance	Dashboards must load within 3-5 seconds. Data extracts or Live queries must be optimized (using indexing or partitioning) to ensure smooth interaction when toggling between "Before" and "After" placement views.

NFR-5	Availability	The visualization must be accessible 24/7 via Tableau Server/Cloud, with scheduled maintenance occurring only during off-peak hours to ensure stakeholders can run reports whenever needed.
NFR-6	Scalability	The solution must support a 50% increase in data volume (new store locations or products) and an increase in concurrent
		users without a degradation in rendering speed.

5.Project Design Phase-II Data Flow Diagram & User Stories

Data Flow Diagrams:

Data Flow Architecture

1. Data Collection & Extraction (The Inbound Layer)

Sources: Raw datasets are gathered from Kaggle, focusing on variables like Product ID, Position, and Sales Volume.

Connection: The systematically gathered data is imported into Tableau Desktop, linking relevant columns to established data fields.

2. Data Preparation (The Processing Layer)

Cleaning & Transformation: Inaccuracies are removed, and raw variables are structured to ensure software compatibility.

Filtering: Analysts refine the scope by focusing on subsets like Demographics or Seasonality to address specific objectives.

3.Data visualization& Dashboard (The Consumption Layer)

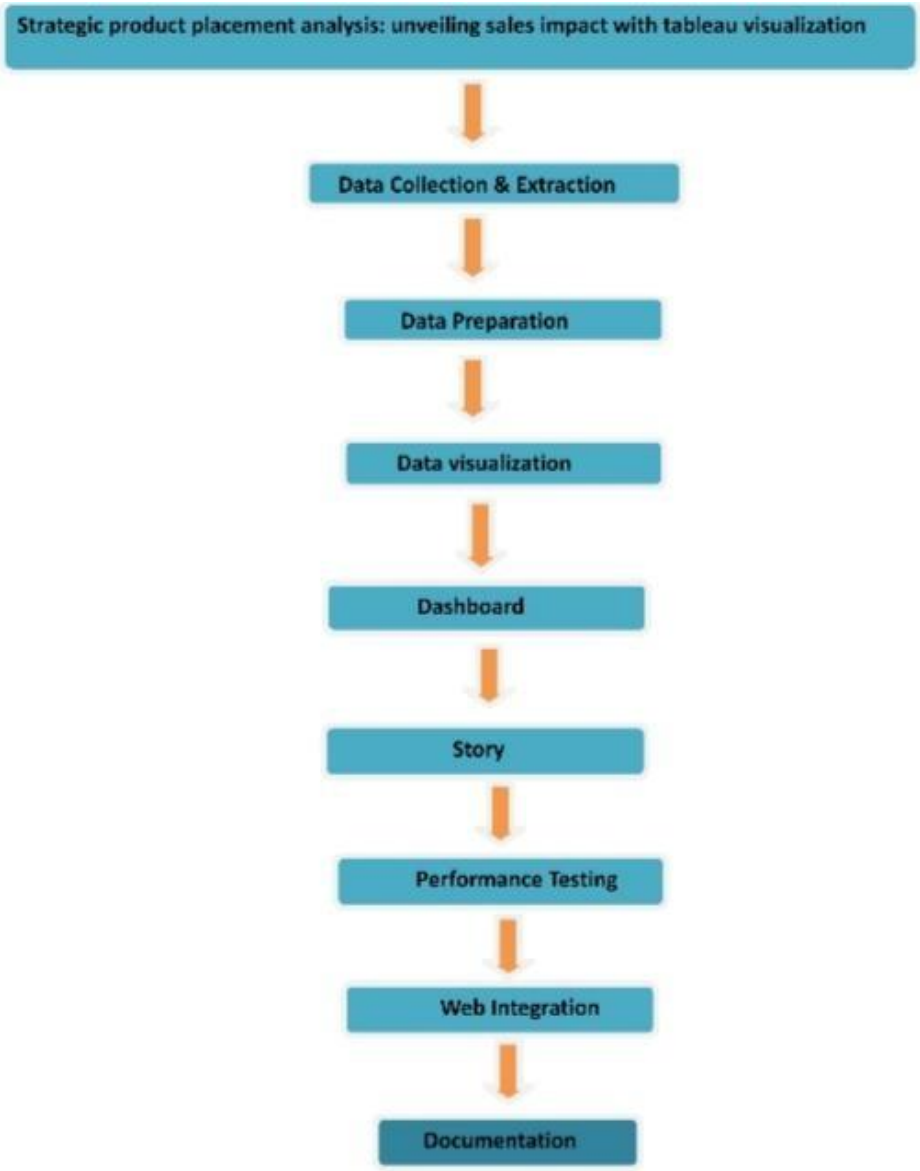
Visual Logic: Processed data flows into unique visualizations (e.g., Avg Sales Volume vs. Product Category) to uncover patterns.
Consolidation: Individual sheets are pulled into a Responsive Dashboard using a floating layout for clear, interactive analysis.

4. Story & Performance Testing (The Insight Layer)

Narrative Flow: The dashboard results move into a Story, breaking the analysis into a sequence of scenes to explain "Sales Impact".
Optimization: Performance testing ensures that Filters and Calculation Fields (like Revenue and ROI) operate efficiently under load.

5. Web Integration & Documentation (The Delivery Layer)

Publishing: The final workbook is saved to Tableau Public, where an embedded code is generated
Deployment: The embedded code is integrated into a Bootstrap template and served via Flask, making the project accessible as a web solution.
Closure: The entire end-to-end flow is recorded in an explanation video and step-by-step documentation.



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Retail Manager	Spatial Dashboard	USN-1	As a manager, I want to see a Heat Map of the store floor.	I can identify "Hot" and "Dead" zones	High	Sprint-1

				visually.		
Retail Manager	Profit Analysis	USN-2	As a manager, I want to compare Eye-level vs. Bottom-shelf profit.	I can see a bar chart showing the margin difference.	High	Sprint-1
Category Analyst	Data Integration	USN-3	As an analyst, I want to upload POS CSV files into the system.	Data is cleaned and mapped to store coordinates.	High	Sprint-1
Category Analyst	Filtering	USN-4	As an analyst, I want to filter sales by Product Category	I can see how "Snacks" perform vs "Dairy" in specific aisles.	Med	Sprint-2
Administrator	Layout Config	USN-5	As an admin, I want to update the store floor plan (X/Y map).	The Tableau map updates to reflect new aisle positions.	Med	Sprint-2

Executive	Reporting	USN-6	As an executive, I want a Weekly Sales Lift report.	I receive an automated email with total revenue impact.	Low	Sprint-2

Project Design Phase-II Technology Stack (Architecture & Stack)

Technical Architecture:

Data Sources:
Integrates unstructured planograms/calendars from Cloud Storage (Google Drive) with structured POS/inventory data from Database Servers.

Preparation:
Uses Tableau to clean SKU inconsistencies and join "Placement" metadata with actual "Sales" figures.

Analytics Engine:
A core modeling layer that calculates specific KPIs like Eye-Level Lift and End-Cap Conversion Rates.

Distribution:
Hosted on Tableau Server/Cloud for security and automated data refreshes.

Accessibility:
Features a Responsive Design optimized for mobile, tablet, and desktop viewing.

User Action:
Enables Category Managers to make rapid "Keep or Move" decisions based on real-time impact visualization.

Core Impact Metrics

To keep the analysis focused, the system prioritizes these three high-level views:

Placement ROI:
Sales growth vs. the cost of premium shelf space.

Heatmap Analysis:
Visualizing "Hot Zones" vs. "Dead Zones" within a store aisle.

A/B Testing:
Direct comparison of sales performance across two different placement strategies

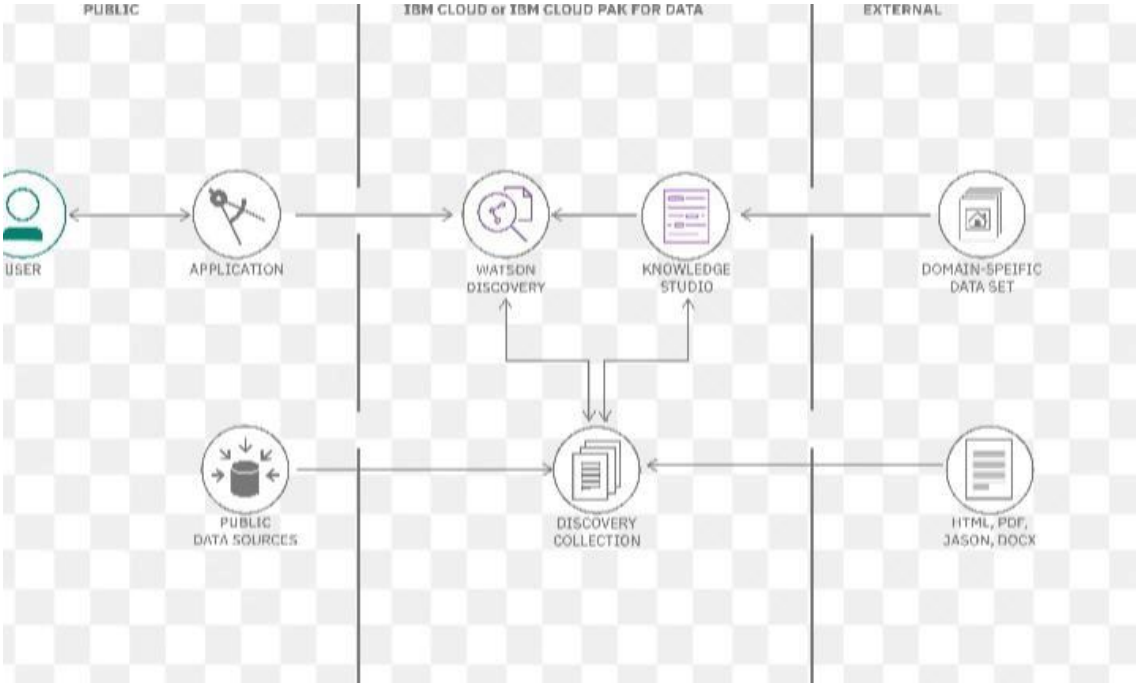


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive Web and Mobile UI for interactive dashboard viewing.	Tableau Web / Mobile App
2.	Application Logic-1	Data harmonization and ETL to merge planograms with sales data.	Tableau Prep / Python

3.	Application Logic-2	Impact modeling to calculate Eye-level vs.	Bottom-shelf sales lift. Tableau Desktop / SQL
4.	Application Logic-3	Automated refresh scheduling for real-time sales trend monitoring.	Tableau Server Scheduler
5.	Database	Stores structured historical POS transactions and store metadata.	MySQL / PostgreSQL
6.	Cloud Database	High-speed warehouse for aggregating multi-store retail datasets.	Snowflake / Google BigQuery
7.	File Storage	Storage for flat-file planograms and manual store audit logs.	Google Drive / Local Filesystem
8.	External API-1	Normalizes sales data based on local weather or seasonal trends.	IBM Weather API
9.	External API-2	Automates user permission syncing and programmatic report delivery.	Tableau REST API
10.	Machine Learning	Forecasts potential revenue lift for proposed shelf-space changes.	Python (Scikit-learn)

11.	Infrastructure	Secure cloud deployment for 24/7 multi-device accessibility.	Tableau Cloud / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Utilizing open-source tools for data transformation and	Python (Pandas/NumPy), PostgreSQL,
		visualization agility.	Open-source API connectors
2.	Security Implementations	Multi-layered security with Row-Level Security (RLS) and encrypted data transit.	SHA-256, SAML/SSO, Tableau RLS, SSL/TLS Encryption.
3.	Scalable Architecture	A 3-tier architecture (Data, Logic, Presentation) designed for multi-device growth.	Micro-services, Kubernetes, Tableau Cloud Clusters.
4.	Availability	High availability through distributed servers and automated health checks.	Load Balancers, AWS Multi-AZ, Tableau Server High Availability.

5	Performance	Optimized query execution through data extracts and caching for fast load times.	Redis Caching, Tableau Data Hyper Extracts, CDN (CloudFront).

6 Project Design Phase Problem – Solution Fit Template Problem – Solution Fit

Retail stores often struggle to understand how product placement (eye-level shelves, end caps, checkout counters, aisle positioning) impacts sales performance. Decisions about placement are frequently based on assumptions rather than data, leading to:

Purpose

- To analyze the impact of product placement on overall sales performance. ● To identify high-performing shelf positions such as eye-level, end caps, and checkout areas.
- To use Tableau for creating interactive dashboards and visual reports.
- To compare sales performance before and after placement changes.
- To optimize store layout for better space utilization and higher revenue.
- To support data-driven decision-making in retail management.
- To improve visibility of high-margin and promotional products.
- To enhance overall profitability through strategic product positioning.



7. **Project Design Phase Proposed Solution Template**

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template

S.No.	Parameter	Description

1	Problem Statement (Problem to be solved)	Retail stores apply various product placement strategies to boost sales but often lack clear insight into their effectiveness. This study analyzes the impact of strategic placement on overall sales performance and uses Tableau to visualize data and identify the most effective strategies for increasing revenue and improving business decisions.
2	Idea / Solution description	This project, Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization, aims to examine how different product placement strategies influence retail sales performance. By analyzing data such as shelf position, promotional displays, product categories, and store locations, the project identifies patterns that affect customer purchasing behavior. Using Tableau
3	Novelty / Uniqueness	This project uniquely combines retail merchandising strategies with advanced data visualization to analyze the impact of product

		placement on sales. Using Tableau, it transforms raw sales data into clear, interactive insights that help retailers optimize store layouts, compare
4	Social Impact / Customer Satisfaction	By analyzing sales patterns using Tableau, retailers can place products in more convenient and visible locations, making shopping faster and more comfortable. This leads to higher customer
5	Business Model (Revenue Model)	By using insights generated from Tableau, businesses can identify high-performing shelf positions and promotional areas to maximize
6	Scalability of the Solution	The solution can be expanded to different retail formats such as supermarkets, malls, and e-commerce platforms. It can also incorporate advanced analytics and predictive modeling in the future, making it adaptable for both small retailers and large retail chains.

8.**Project Design Phase Solution Architecture**

Solution Architecture:

Data Sources Layer

- Point of Sale (POS) Systems – Transaction-level sales data (product, quantity, revenue, date, store).
- Inventory Database – Stock levels, product categories, shelf allocation. ● Store Layout Data – Product placement zones (entrance, aisle-end, eye-level shelves, checkout counters).
- Promotional Data – Discounts, campaigns, in-store offers.

Data Integration & Preparation Layer

- Data Extraction – Connect Tableau to SQL databases, Excel files, or cloud storage.
- Data Cleaning – Remove duplicates, standardize product names, handle missing values.
- Data Transformation –
- Create calculated fields (Sales Growth %, Average Basket Size, Placement Impact Score).
- Merge sales data with placement zones.
- Data Aggregation – Summarize by product, category, placement zone, and time period.

Data Storage Layer

- Centralized Data Warehouse / Data Mart ● Structured tables for:
- Sales Facts
- Product Dimensions
- Store Dimensions
- Placement Dimensions

Analytics & Business Logic Layer

- KPI Calculations:
- Total Sales
- Sales by Placement Zone
- Conversion Rate
- Revenue per Square Foot
- Before vs After Placement Comparison
- Trend Analysis (Weekly / Monthly / Seasonal)
- Correlation analysis between placement position and sales performance

Visualization Layer (Using Tableau)
Interactive dashboards built in Tableau:

- Sales by Placement Heatmap
- Trend Analysis Line Charts
- Product Category Performance Dashboard
- Store-wise Placement Impact Comparison
- Shelf Position vs Revenue Bar Charts

User Layer

- Retail Managers
- Category Managers
- Marketing Team
- Store Operations Team
- Business Analysts

Deployment & Maintenance

- Publish dashboards to Tableau Server / Tableau Cloud
- Schedule automatic data refresh
- Role-based access control
- Performance monitoring and periodic optimization

9. **Project Planning Phase Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)**

Product Backlog, Sprint Schedule, and Estimation (4 Marks)
Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection and Extraction of Data	USN-1	1.Collect the dataset 2.Understanding the Data 3.Connecting the	3	High	Bala Mahalakshmi Konakalla

			dataset with Tableau			
Sprint-1	Data Preparation	USN-2	Prepare the Data for Visualization	5	High	Chenikala Nagendra
Sprint-2	Data Visualization	USN-3	No of Unique Visualizations	4	Low	Bala Mahalakshmi Konakalla
Sprint-1	Dashboard	USN-4	Responsive and Design of Dashboard	3	Medium	Kusuma Leela Pittala
Sprint-1	Story	USN-5	No of Scenes of Story	5	High	Mamidi Durga Ambika
Sprint-2	Performance Testing	USN-6	1.Utilization of Filters 2. No of Calculation Fields 3. No of Visualizations/ Graphs	8	High	Hazara Shaik
Sprint-3	Web integration	USN-7	Dashboard and Story embed with UI With Flask	5	High	Mamidi Durga Ambika

Sprint-4	Project Demonstration & Documentation	USN-8	1.Record an explanation Video for the project end to end solution 2 Project Documentation-Step by step project development	4	High	Bala Mahalakshmi Konakalla
			procedure			

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	**Total Story Points**	**Duration**	**Sprint start date**	**Sprint end Date (planned)**	**Story points completed(as on planned end Date)**	**Sprint release date (actual)**
Sprint-1	Data Collection and Extraction of Data	6 days	4/2/2026	10	20	4/2/2026
Sprint-2	Data Preparation	3 days	10/2/2026	13	20	10/2/2026
Sprint-3	Data Visualization	3days	13/2/2026	16	20	13/2/2026
Sprint-4	Dashboard	2 days	16/2/2026	18	20	16/2/2026

Sprint-5	Story	2 days	19/2/2026	21	20	19/2/2026
Sprint-6	Performance Testing	2days	21/2/2026	22	20	21/2/2026
Sprint-7	Web integration	2days	22/2/2026	23	20	22/2/2026
Sprint-8	Project Demonstration & Documentation	4 days	23/2/2026	27	20	23/2/2026

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

10.Project Development Phase

Model Performance Test

Model Performance Testing:

Project team shall fill the following information in model performance testing template

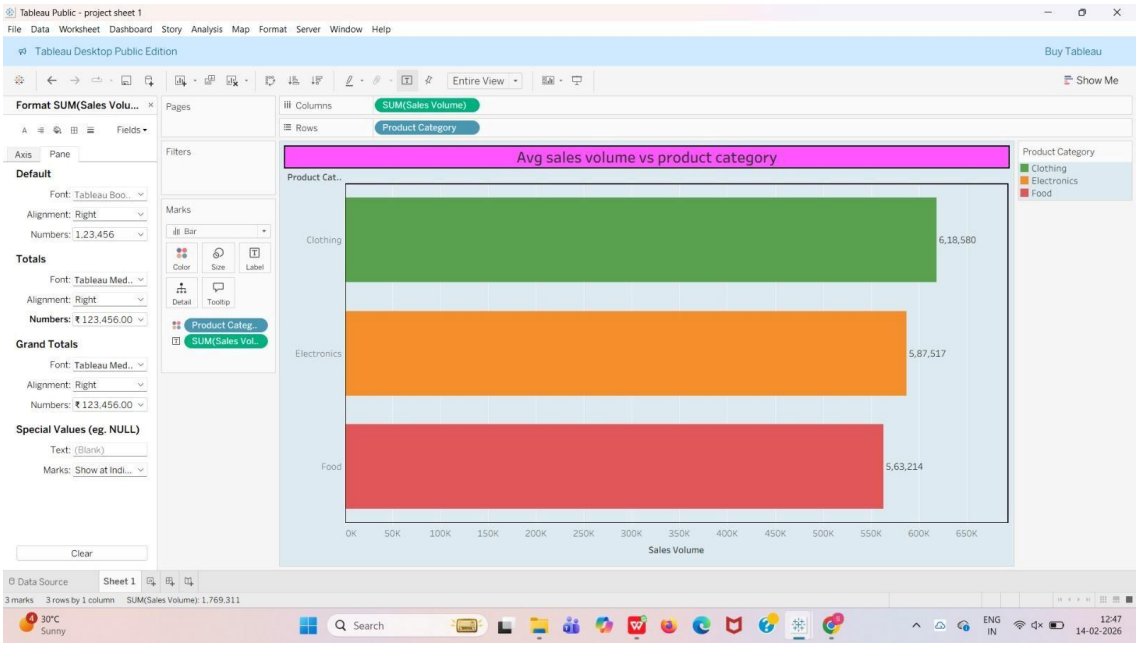
S.No	Parameter	Screen shot / values

1	Data Rendered	Displayed sales performance data across store zones, product categories, and promotion periods. Shows rendered Heat Map (Store Hot Zones), Scatter Plot (Correlation between shelf height and sales), Side-by-Side Bar Chart (Placement A vs. Placement B), and KPI Cards (Total Revenue, Conversion Rate, Placement Lift %).
2	Data Preprocessing	Cleaned transactional records, handled missing store IDs, standardized currency formats, and converted inventory units. Created a Product Hierarchy (Department > Category > SKU) and Location Hierarchy (Region > Store > Aisle >
		Shelf).
3	Utilization of Filter	Filters applied: Store Location, Product Category, Time Frame (Seasonal/Holiday), Placement Type (End-cap, Eye-level, Floor stack), and Customer Loyalty Tier. Enables users to isolate the impact of specific display changes on sales volume.
4	Calculation Fields Used	Sales Lift (%) = \frac{\text{Sales (Strategic)} - \text{Sales (Standard)}}{\text{Sales (Standard)}} Conversion Rate = \frac{\text{Transactions}}{\text{Footfall Count}} Revenue Velocity = \frac{\text{Total Revenue}}{\text{Days on Display}}

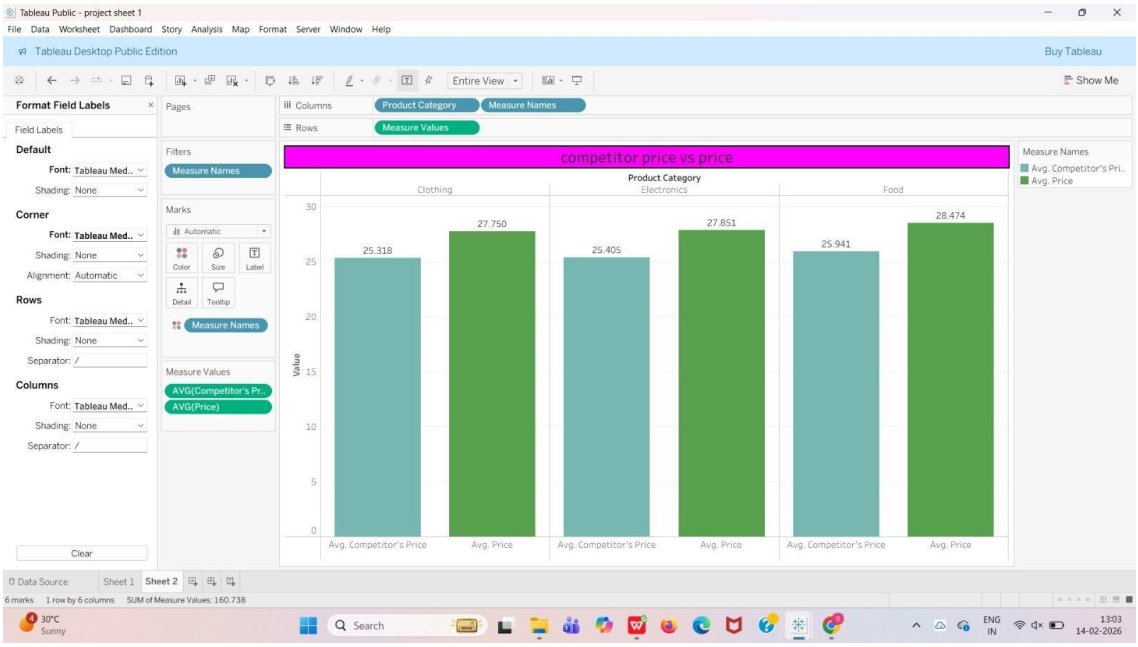
5	Dashboard Design	No. of Visualizations: (Heat Map – Foot traffic vs. Sales; Scatter Plot – Shelf position impact; Bar Chart – Category comparison; Highlight Table – Weekday performance; KPI Cards – Revenue & Lift metrics).
6	Story Design	Executive Summary, Aisle Performance Breakdown, Product Visibility Impact, Promotional Lift Analysis, and Space Optimization Recommendations.

Data Visualization

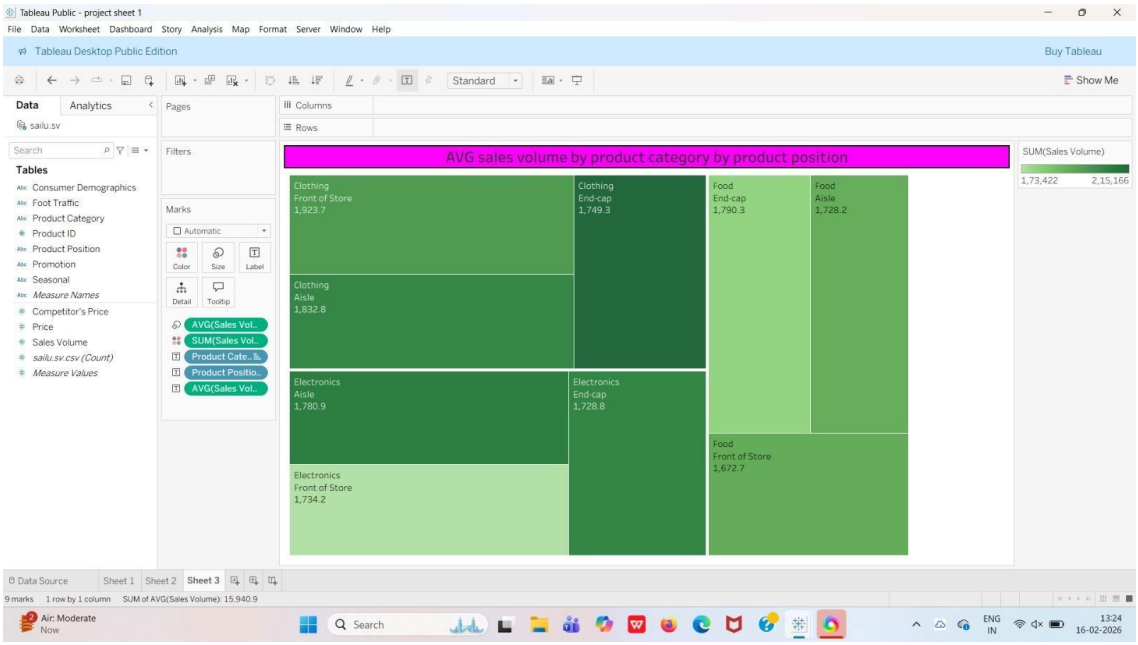
Activity 1.1: Avg Sales Volume vs Product Category



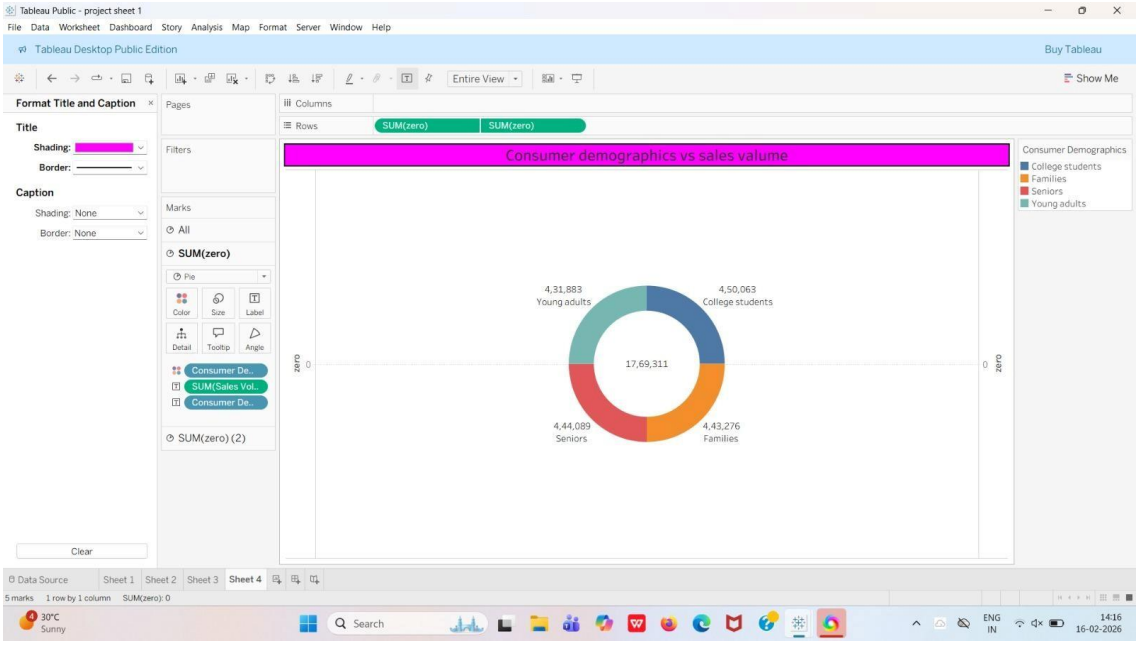
Activity 1.2: Competitor Price Vs Price



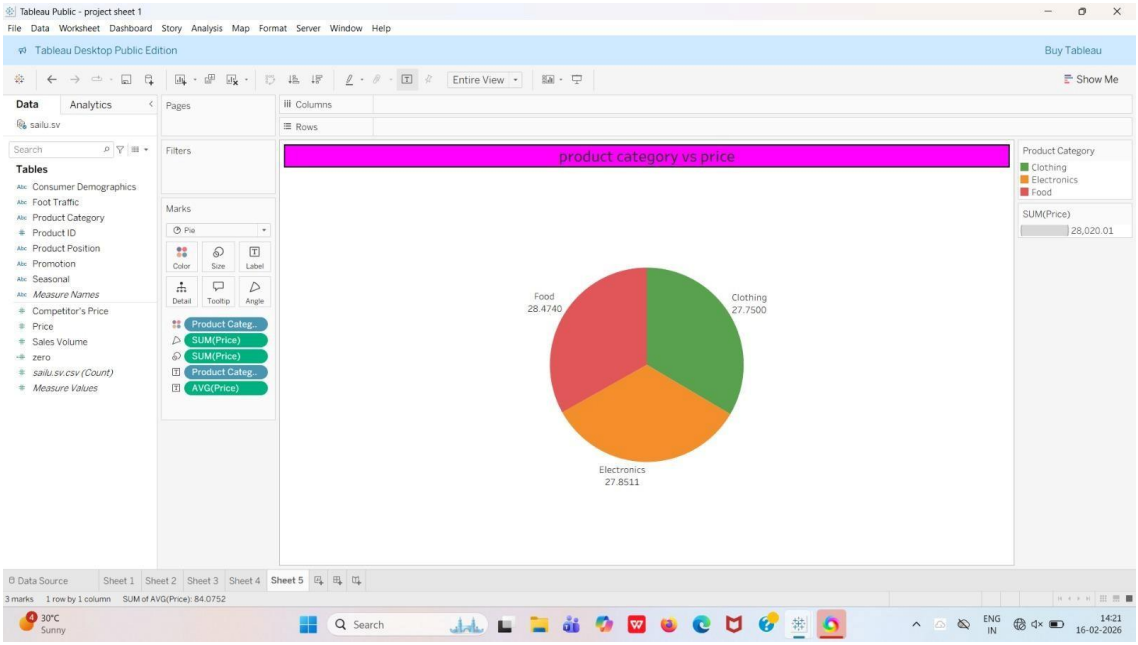
Activity 1.3: Avg Sales Volume by Product Category by Product Position



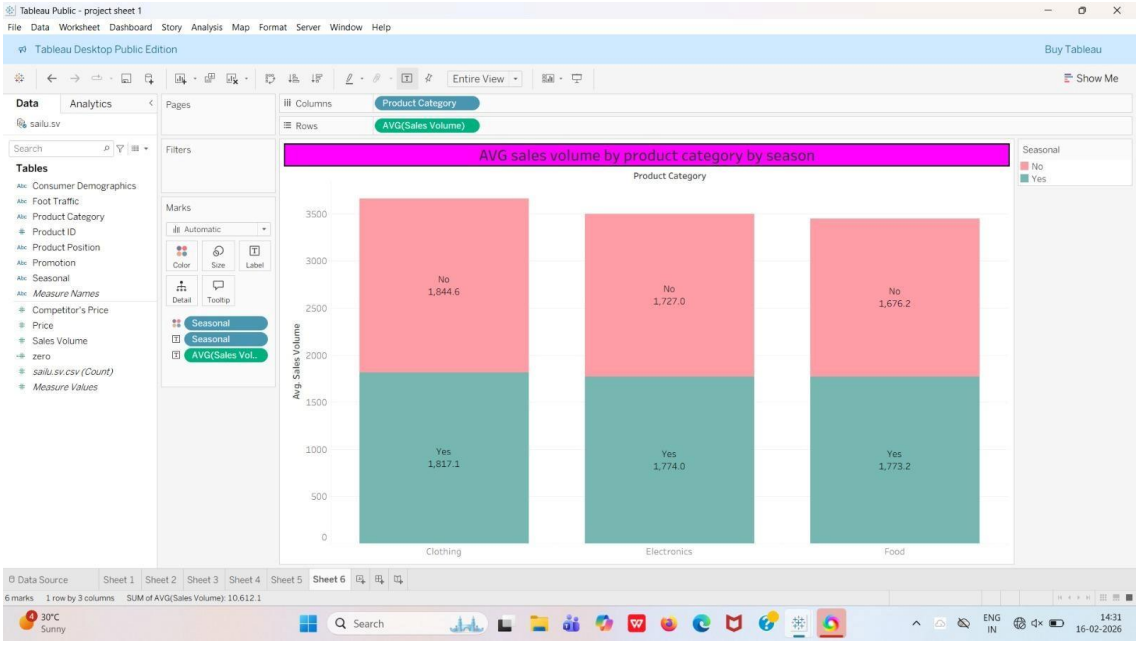
Activity 1.4: Consumer Demographics vs Sales Volume



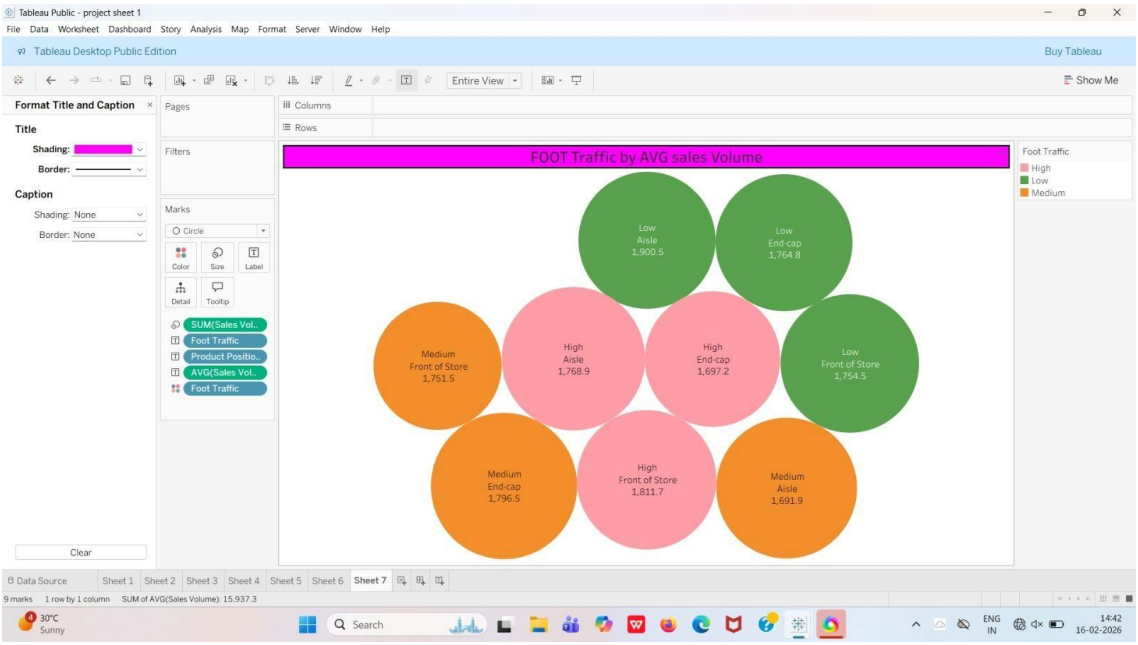
Activity 1.5: Product Category vs Price



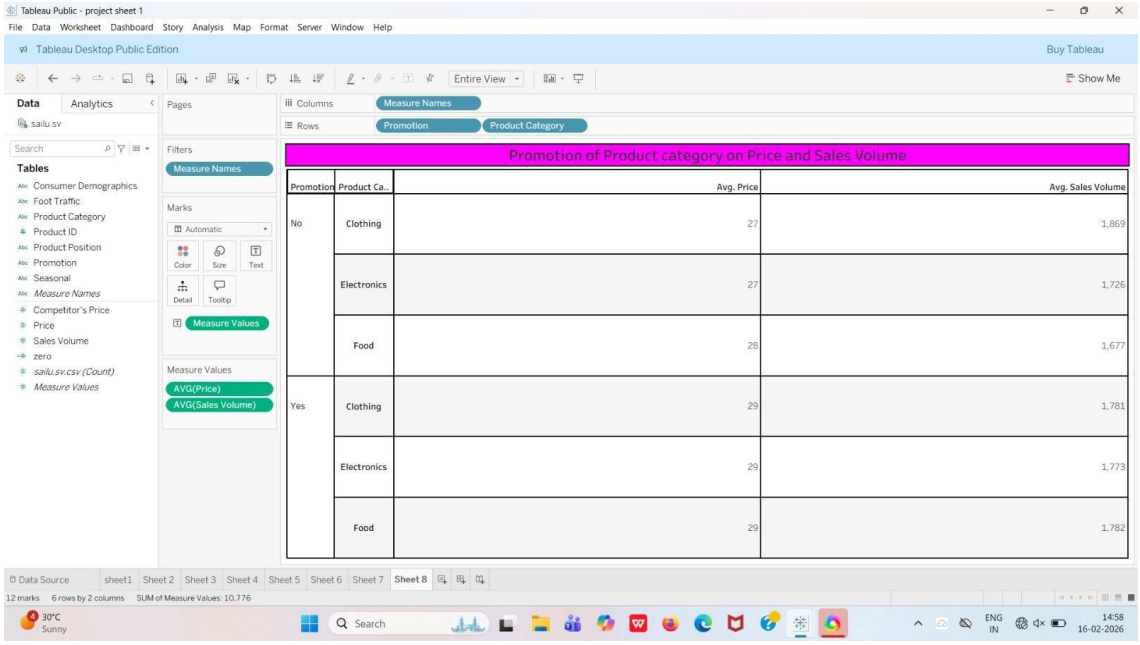
Activity 1.6: Avg Sales Volume by Product Category by Season



Activity 1.7: Foot Traffic by Avg Sales Volume

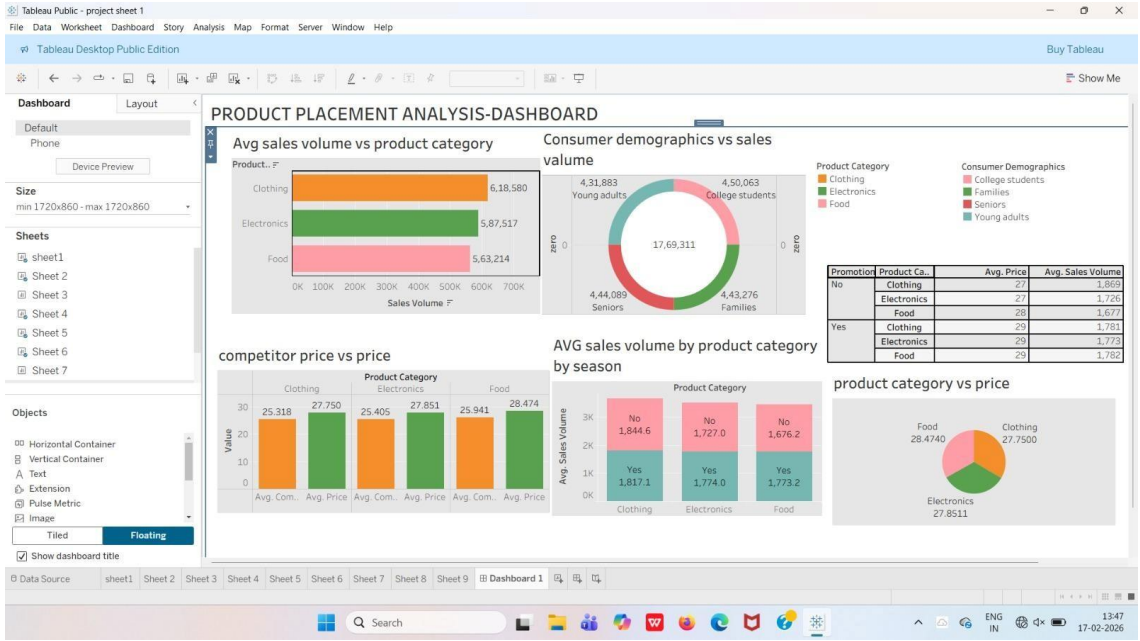


Activity 1.8: Promotion of Product Category on Price and Sales Volume



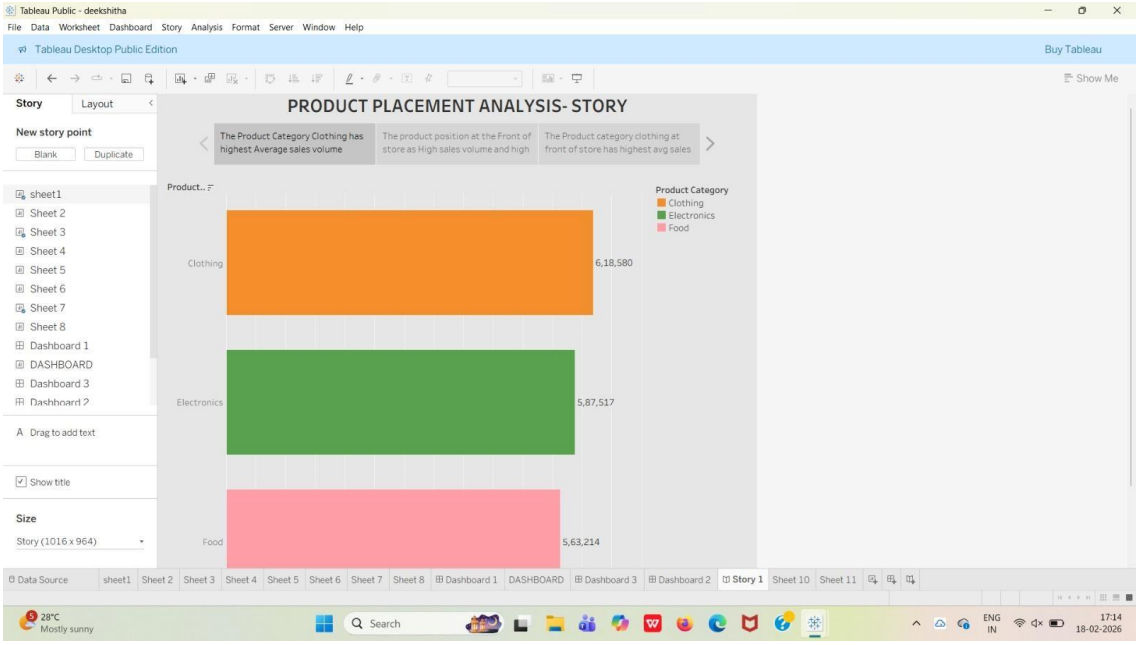
Dashboard:

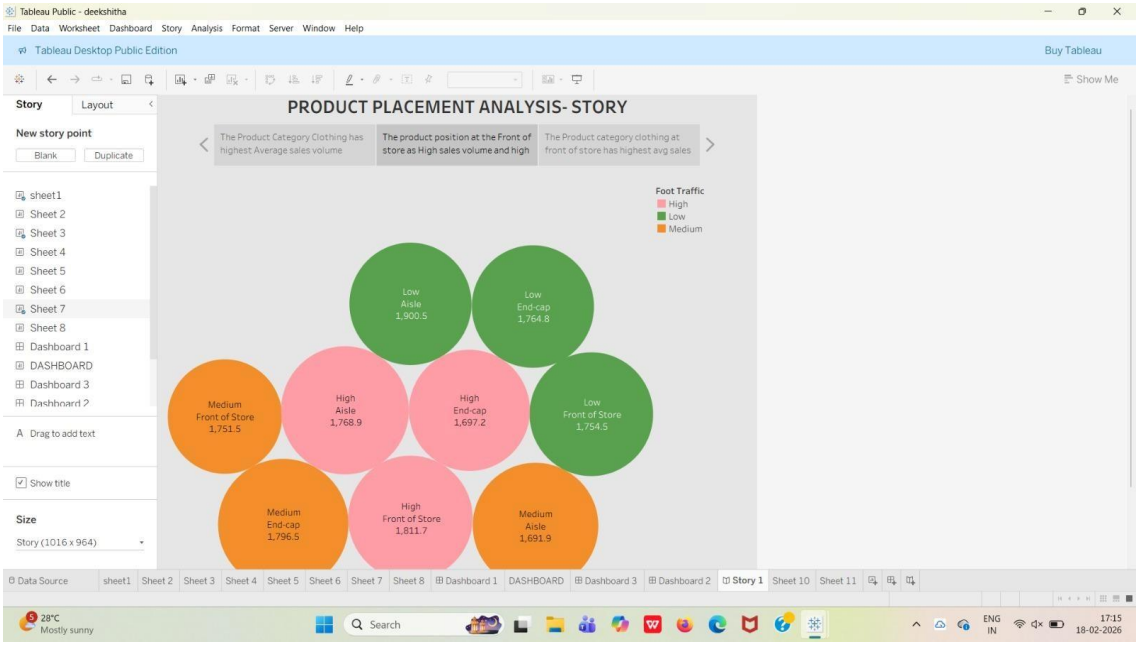
Dashboard: Product Placement Analysis – Dashboard



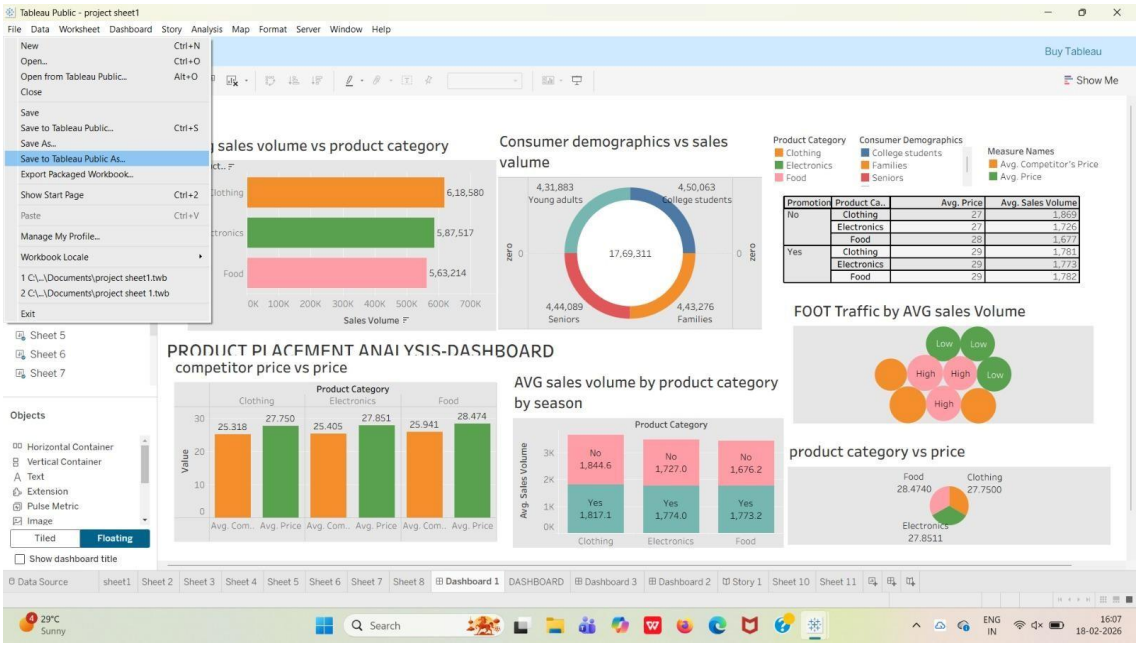
5) Story

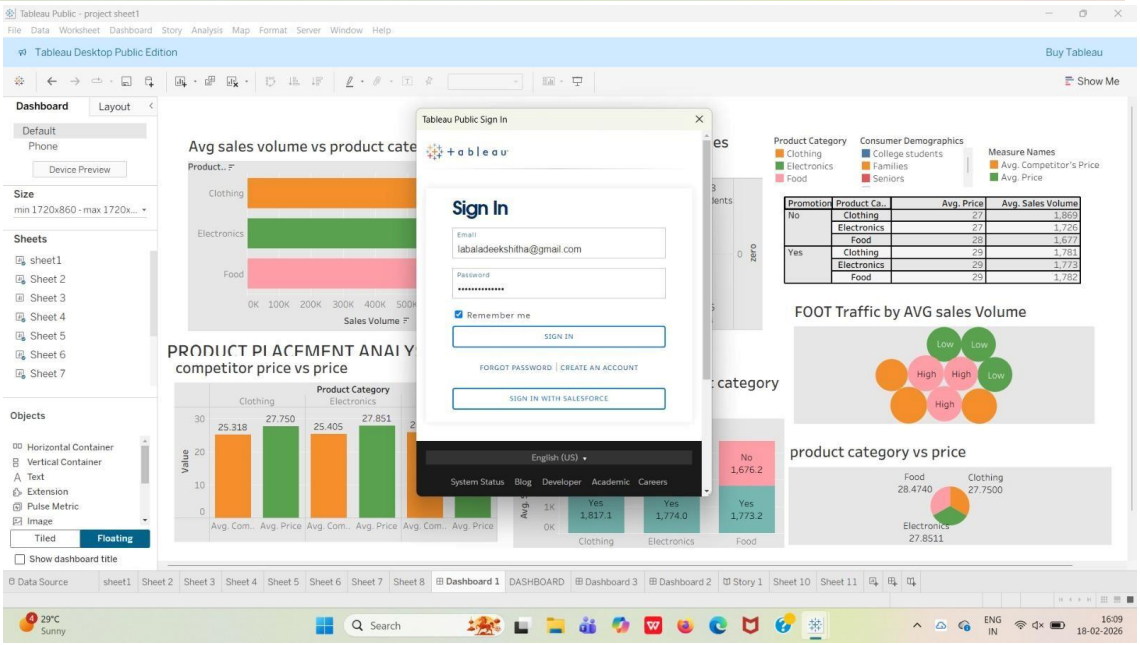
Product Placement Analysis – Story

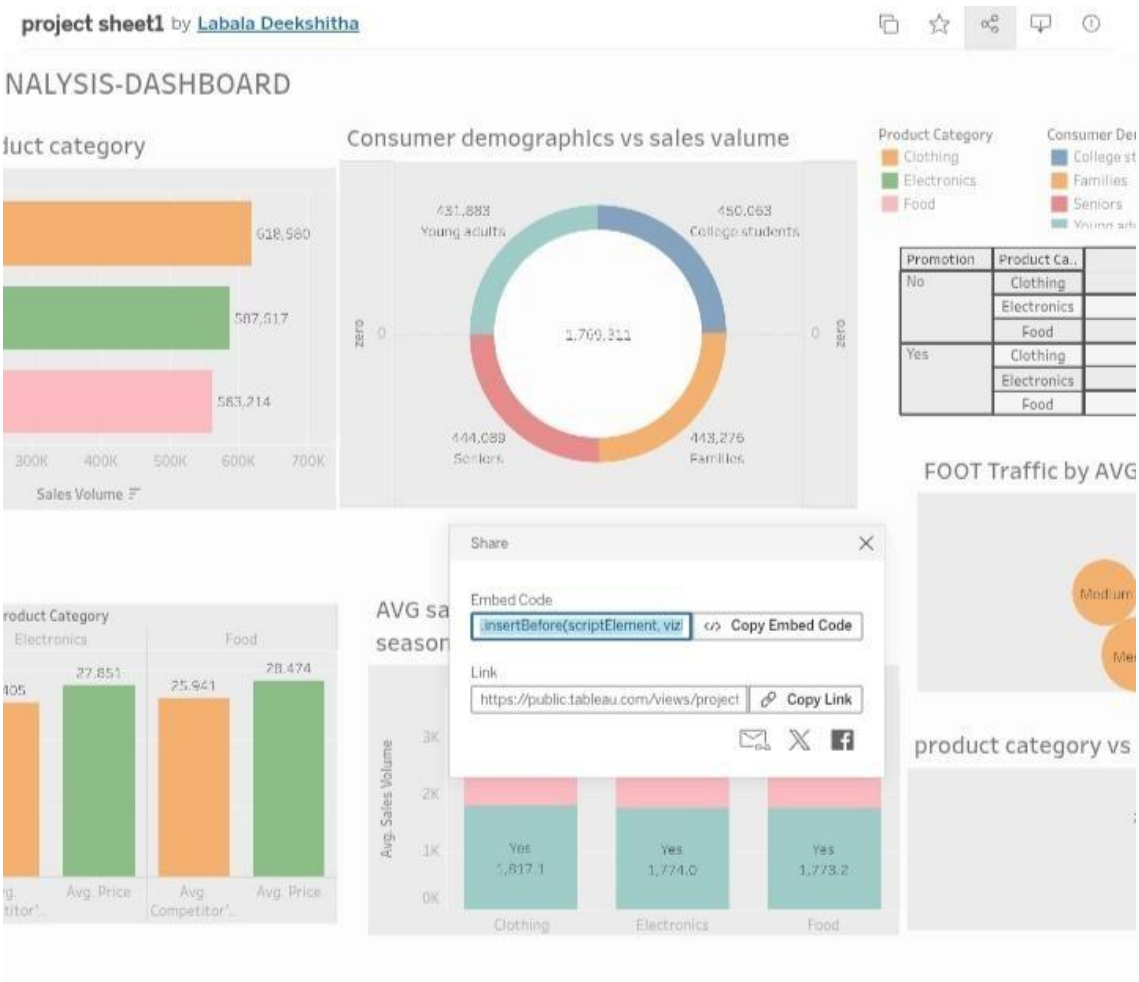

Tableau Public - deekshitha
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Drag to add text
Show title
Size
Story (1016 x 964)
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high
The Product category clothing at front of store has highest avg sales
Product..
Product Category
Clothing
Electronics
Food
Clothing
6,18,580
Electronics
5,87,517
Food
5,63,214
Data Source
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Story 1
Sheet 10
Sheet 11
28°C
Mostly sunny
Search
ENG
IN
17:14
18-02-2026

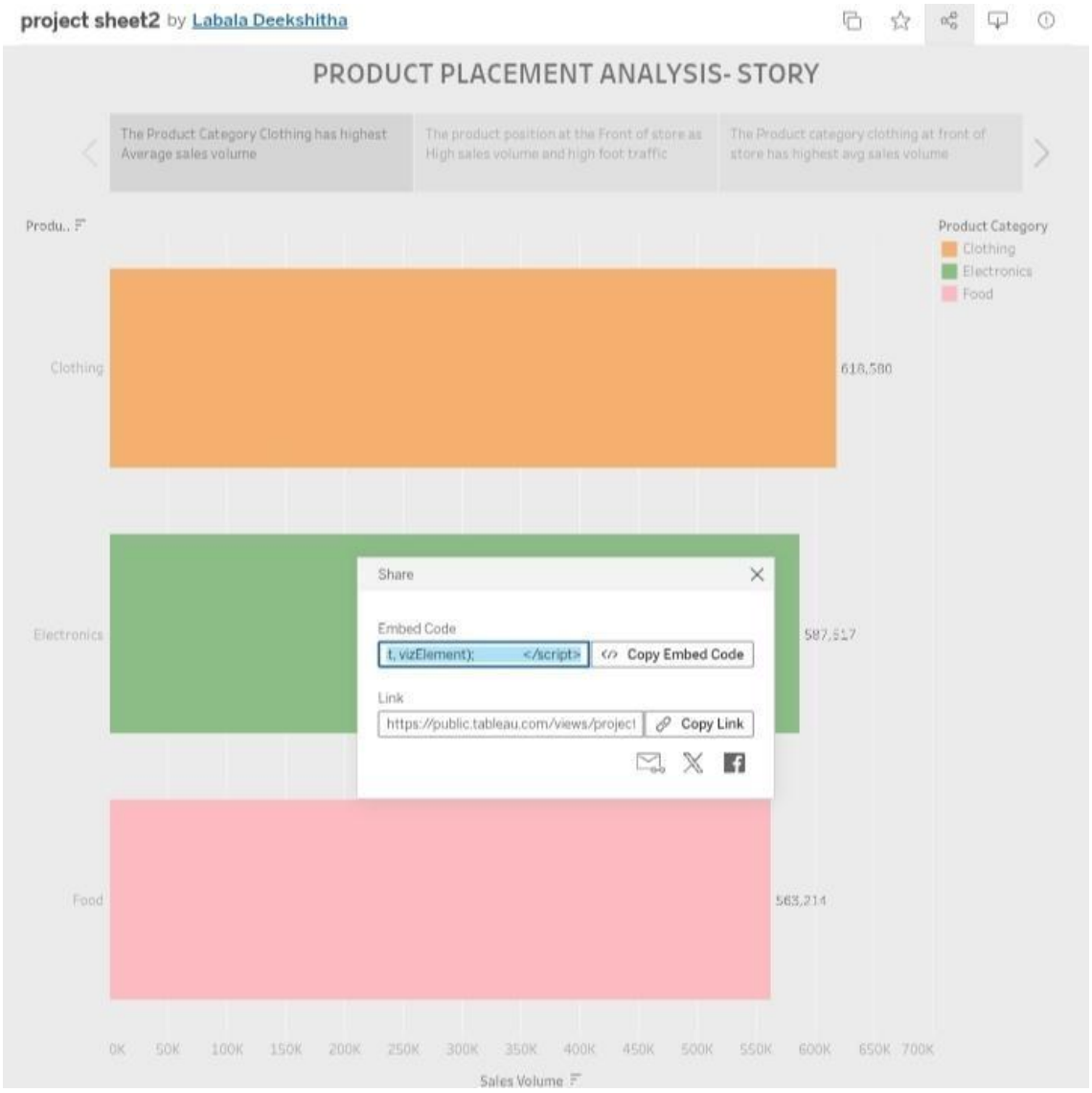

Tableau Public - deekshitha
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Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Drag to add text
Show title
Size
Story (1016 x 964)
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high
The Product category clothing at front of store has highest avg sales
Foot Traffic
High
Low
Medium
Low Aisle 1,900.5
Low End-cap 1,764.8
Medium Front of Store 1,751.5
High Aisle 1,768.9
High End-cap 1,697.2
Low Front of Store 1,754.5
Medium End-cap 1,796.5
High Front of Store 1,811.7
Medium Aisle 1,691.9
Data Source
sheet1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Dashboard 1
DASHBOARD
Dashboard 3
Dashboard 2
Story 1
Sheet 10
Sheet 11
28°C
Mostly sunny
Search
ENG
IN
17:15
18-02-2026

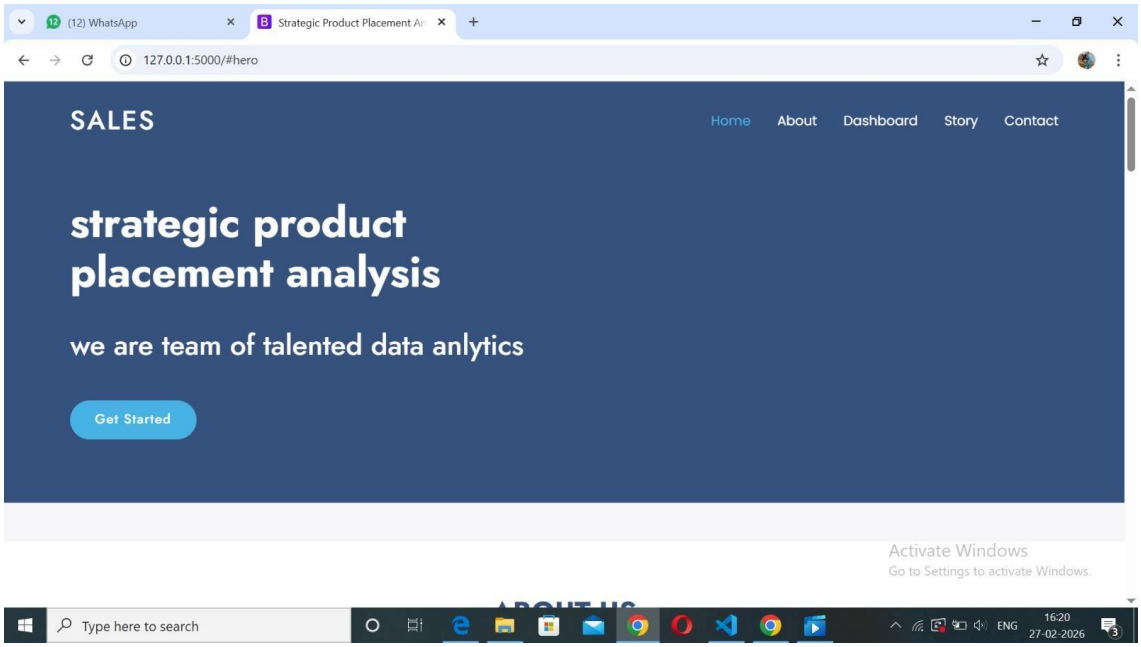
Web integration

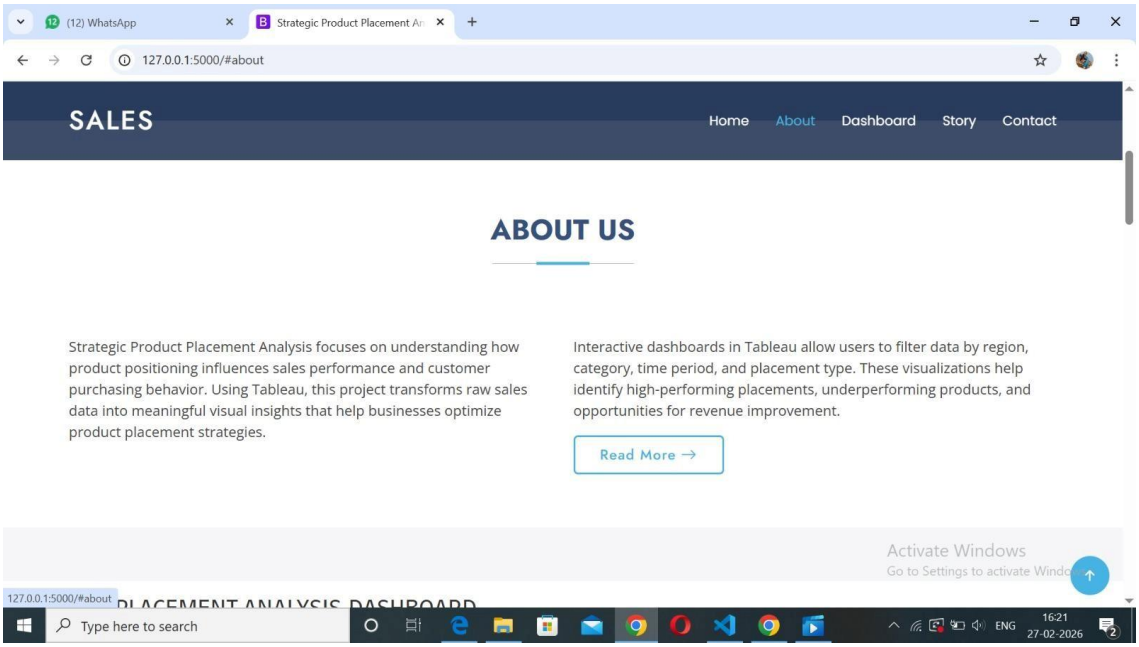


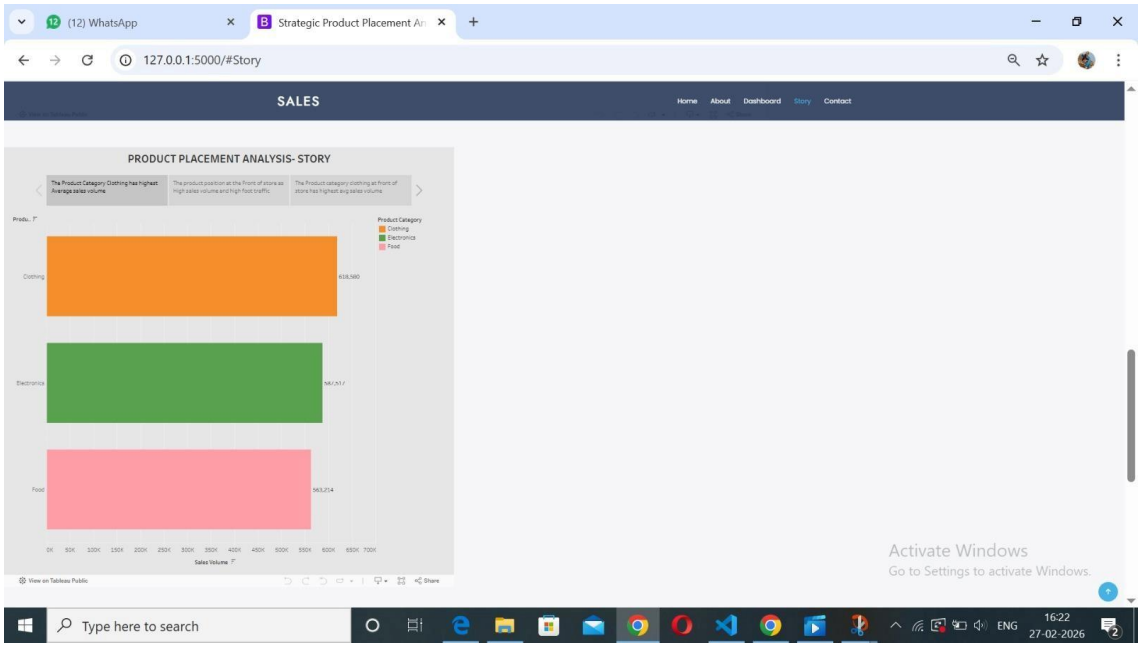


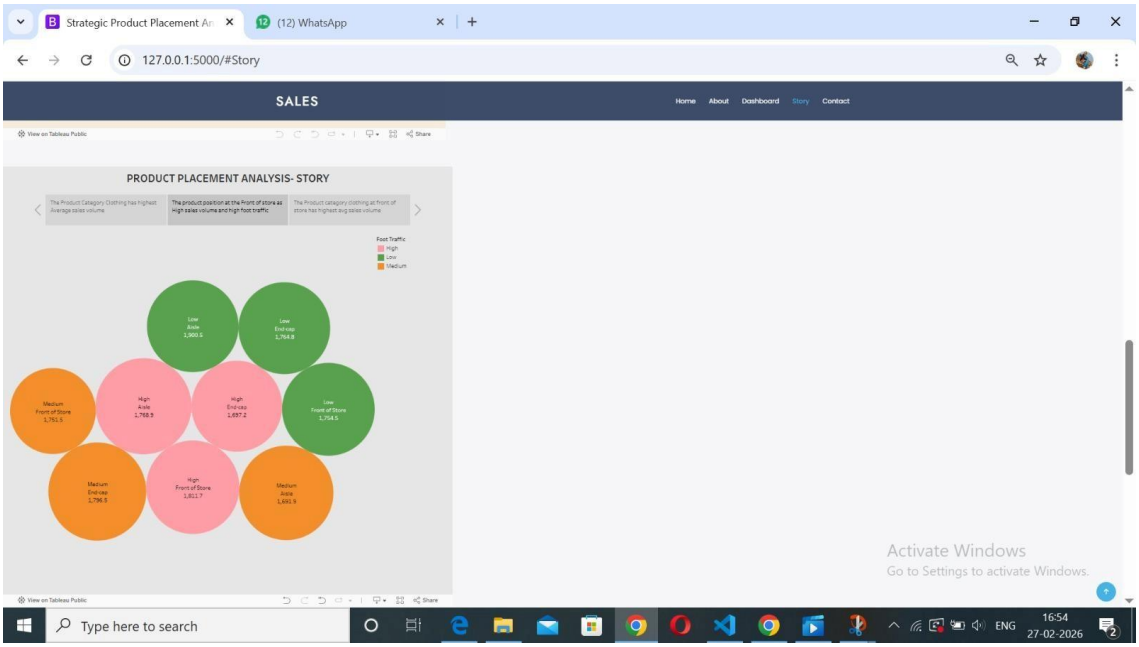

project sheet1 by Labala Deekshitha
NALYSIS-DASHBOARD
luct category
618,580
587,517
583,214
300K
400K
500K
600K
700K
Sales Volume
Consumer demographics vs sales valume
431,883
Young adults
450,063
College students
1,769,311
444,089
Seniors
443,276
Families
zero
0
Product Category
Clothing
Electronics
Food
Consumer De
College st
Families
Seniors
Promotion
Product Ca..
No
Clothing
Electronics
Food
Yes
Clothing
Electronics
Food
FOOT Traffic by AVG
Medium
roduct Category
Electronics
Food
405
27,851
25,941
28,474
Avg. Price
Avg. Competitor'...
Avg. Price
Share
Embed Code
insertBefore(scriptElement, viz
Copy Embed Code
Link
https://public.tableau.com/views/project
Copy Link
AVG sa
season
Avg. Sales Volume
3K
2K
1K
0K
Yes
1,817.1
Yes
1,774.0
Yes
1,773.2
Clothing
Electronics
Food
product category vs

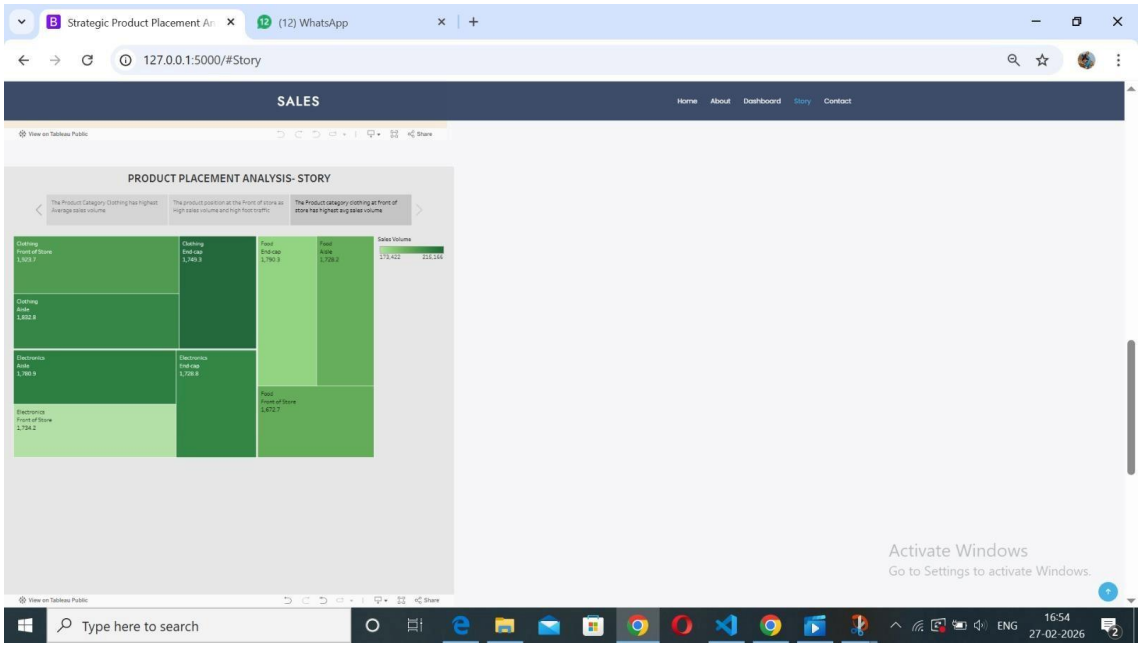

project sheet2 by Labala Deekshitha
PRODUCT PLACEMENT ANALYSIS- STORY
The Product Category Clothing has highest Average sales volume
The product position at the Front of store as High sales volume and high foot traffic
The Product category clothing at front of store has highest avg sales volume
Produ..
Product Category
Clothing
Electronics
Food
Clothing
616,580
Electronics
587,517
Food
583,214
Share
Embed Code
t, vizElement); </script>
Copy Embed Code
Link
https://public.tableau.com/views/project
Copy Link
0K 50K 100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K 650K 700K
Sales Volume

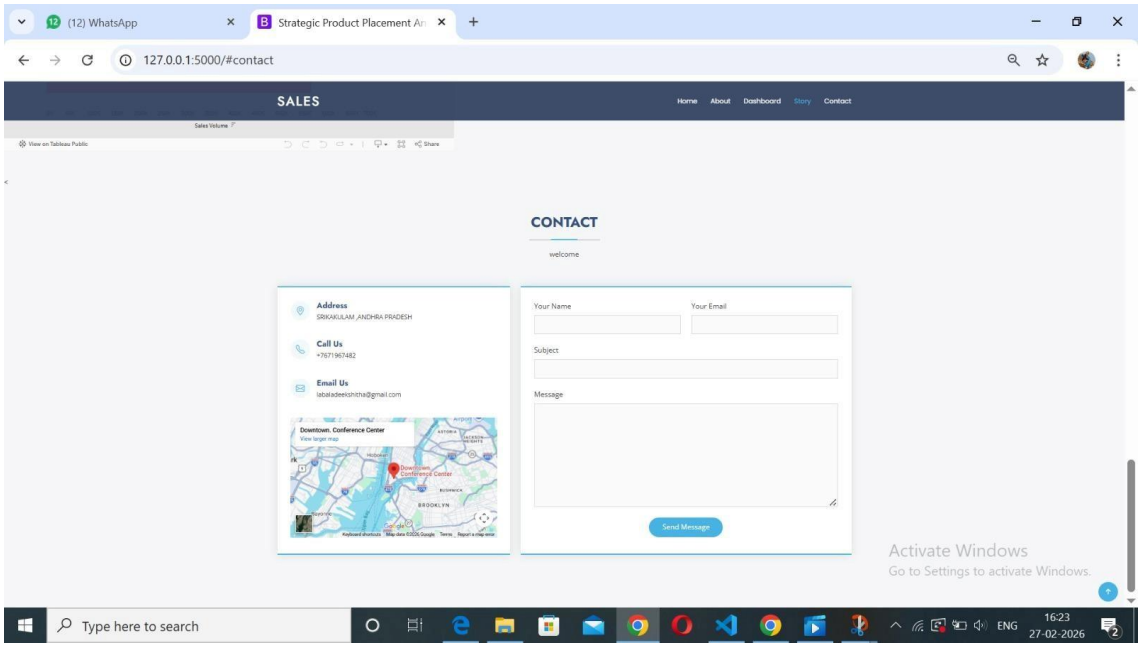

(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#hero
SALES
Home
About
Dashboard
Story
Contact
strategic product placement analysis
we are team of talented data anlytics
Get Started
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:20
27-02-2026


(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#about
SALES
Home
About
Dashboard
Story
Contact
ABOUT US
Strategic Product Placement Analysis focuses on understanding how product positioning influences sales performance and customer purchasing behavior. Using Tableau, this project transforms raw sales data into meaningful visual insights that help businesses optimize product placement strategies.
Interactive dashboards in Tableau allow users to filter data by region, category, time period, and placement type. These visualizations help identify high-performing placements, underperforming products, and opportunities for revenue improvement.
Read More →
Activate Windows
Type here to search
ENG
16:21
27-02-2026


(12) WhatsApp
Strategic Product Placement An
127.0.0.1:5000/#Story
SALES
Home
About
Dashboard
Story
Contact
PRODUCT PLACEMENT ANALYSIS- STORY
Clothing
Electronics
Food
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:22
27-02-2026


Strategic Product Placement An
(12) WhatsApp
127.0.0.1:5000/#Story
SALES
Home
About
Dashboard
Story
Contact
PRODUCT PLACEMENT ANALYSIS- STORY
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
16:54
27-02-2026





ADVANTAGES AND DISADVANTAGES

Advantages

Data-Driven ROI: Replaces guesswork with exact "Sales Lift" metrics for every shelf position. Visual Clarity: Uses Tableau heatmaps to instantly identify high-traffic "Hot Zones" and underperforming "Cold Zones".

Unified Insights: Merges siloed data from SQL databases and Google Drive into one accessible dashboard.

Real-Time Agility: Enables store managers to make immediate layout adjustments based on live performance data.
Consumer Understanding: Links demographics (Families, Seniors) to specific placement success for targeted marketing.

Disadvantages

Data Complexity: Requires significant effort in the "Data Preparation" phase to clean and transform raw POS records.
Integration Hurdles: Initial setup to connect Google Drive planograms with SQL databases can be technically challenging.
Maintenance Needs: Dashboards require regular updates to account for seasonal product changes and new SKUs.
High Initial Effort: Sprint-1 and Sprint-2 require heavy lifting from team members before any visual insights are produced.
Human Bias: Even with data, physical store constraints (e.g., fixed heavy shelving) may limit the ability to move products to "ideal" spots.

Conclusion

The Strategic Product Placement Analysis project successfully demonstrates how data-driven visualization can transform traditional retail management into a high-precision science. By bridging the gap between raw sales figures and physical store geography, the project provides a comprehensive solution for "unveiling" the hidden drivers of consumer behavior.

FUTURE SCOPE
Predictive AI Integration: Incorporating Machine Learning models to predict "Sales Lift" before a physical layout change is even made.

Real-Time IoT Sensors: Using in-store shelf sensors to update Tableau heatmaps instantly as products are picked up, rather than waiting for POS sync.
AR Visualization: Developing Augmented Reality (AR) tools for store managers to overlay Tableau performance metrics directly onto physical shelves using mobile devices.
Advanced Demographic Filtering: Integrating loyalty card data to track individual "Shopper Journeys" beyond broad categories like "Families" or "Seniors".
Automated Reordering: Linking shelf-level performance data to supply chain systems to automatically trigger restocks for high-velocity "Hot Zones".
Global Store Benchmarking: Expanding the Flask UI to compare placement effectiveness across multiple geographic locations or store formats.

APPENDIX
Source Code(if any)
Flask Folder link:

https://drive.google.com/drive/folders/15abhxtDxY48c_5iDZz62ENd9pR4aT33l?usp=sharin g

Dataset link

https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales

Dashboard Public Link:

https://public.tableau.com/views/projectsheet1_17714887574540/Dashboard1?:language=enUS&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link Story Public Link:

https://public.tableau.com/views/projectsheet2_17714891134110/Story1?:language=en-US& publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link